Interacting Random-field Dipole Defect Model for Heating in Semiconductor-based Qubit Devices

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Semiconductor qubit devices suffer from the drift of important device parameters as they are operated. The most important example is a shift in qubit operating frequencies. This effect appears to be directly related to the heating of the system as gate operations are applied. We show that the main features of this phenomenon can be explained by the two-level systems that can also produce charge noise, if these systems are considered to form an interacting random-field glass. The most striking feature of the theory is that the frequency shift can be non-monotonic in temperature. The success of the theory narrows considerably the possible models for the two-level systems.

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I. INTRODUCTION

There has been considerable progress in semiconductor quantum computing, with significant strides in scaling up and in gate fidelities [1, 2]. The chief difficulty is decoherence, with charge noise as the main culprit [3–11]. The precise nature of the two-level systems (TLS) that produce the noise remains elusive. There have been extensive characterizations of the noise spectrum [12, 13], and the spatial correlations in the noise [14, 15] in different devices. These observations can help in the elucidation of the nature of the TLS, but the data are not currently sufficient to pin things down precisely.

Another problem, at first sight quite separate, that interferes with qubit operation is the pulse-induced resonance shift (PIRS). This is a shift in the operating frequency of the qubits as a computation proceeds. This is highly problematic, since continual recalibration of the system is not practical. Quadrature control [16] and prepulsing [17] can mitigate but not eliminate this issue. PIRS has recently received an intensive experimental study [18].

Here we propose that the source of PIRS is also a group of TLS, perhaps the same group that gives charge noise. Hence detailed observations of PIRS may provide additional insight into the microscopic origin of the noise. We proceed in the time-honored fashion of proposing a phenomenological model that explains the data, and then seeing what constraints the model places on the underlying physics of the system.

Let us take a concrete situation in which the system is in a resting state at low temperature T for times $t < 0$ and the operations, which involve microwave pulses that feed energy into the system, begin at $t = 0$ and end at some later time t_f. PIRS is a time-dependent shift $\Delta f(t)$ with $\Delta f(t = 0) = 0$ by definition. Δf is a function of time that ultimately reverts to the base state some time after the operations have ceased. PIRS appears to be rather ubiquitous in semiconductor qubits, but there is considerable variability in how it manifests itself. Early observations found positive shifts ($\Delta f \geq 0$) of order a few MHz [17]. The magnitude of the shift was an increasing function of the energy injected by the pulses. It also depended on the details of the electron wavefunctions in the dots, for example on the dot occupations. The MHz magnitude of the shifts is fairly typical for quantum dot qubits. Importantly, Δf can also be negative [16, 19]. The decay time after t_f varies in the dot systems, with values from 0.5 ms [1] to 38 μs [19] having been observed. PIRS also occurs in donor-based qubit systems [20], though $|\Delta f|$ is much smaller, of order 10s of kHz. Effects of a similar magnitude are seen in flip-flop qubits [21]. In this work we concentrate on experiments in dots, but we expect the theory to apply more broadly.

Our interpretation is based on the fact that dot and donor systems share the feature that the qubit operating frequency depends on the spatial position of the qubit. The arbitrary sign of Δf then suggests that a change in the electric field on the qubit is the origin of PIRS. Experimentally, it now appears to be clear that PIRS is essentially a thermal heating effect rather than a mechanical effect [1, 18]. This is also supported by the characteristic return to a base state, most naturally interpreted as a return to thermal equilibrium. The most striking feature of the results is that the magnitude of Δf is typically not monotonic in temperature (T), instead rising to a maximum at about 200-300 mK, then decreasing [18].

II. MODEL

To explain these observations, we introduce a model based on the charged TLS that are known to exist in these devices, and that in fact are also responsible, at least in part, for the decoherence of the qubits. These charged defects or traps are modeled as a collection of N fluctuating electric dipoles. The jth dipole fluctuates between states $s_j \mathbf{p}_j$, where $s_j = \pm 1$ and $\mathbf{p}_j$ is a fixed vector for each j. For simplicity we assume that the dipoles all have the same magnitude: $|\mathbf{p}_j| = p_0$. This is reasonable if all the TLS have the same physical origin. The dipoles can have a non-zero equilibrium moment which is random in direction and they interact via the long-

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range Coulomb interaction. We call this the Interacting Random-field Glass Model (IRGM). Somewhat similar models have been introduced to understand charge noise (rather than equilibrium fields) in dot systems [22] and also in the context of superconducting qubit systems to explain fluctuations in the relaxation time T_1 [23, 24].

The electric field $\langle \mathbf{F}_q \rangle$ on a qubit at the origin of coordinates is

$$\langle \mathbf{F}_q \rangle = \frac{1}{4\pi\varepsilon} \sum_{j=1}^{N} \langle s_j \rangle \frac{3(\mathbf{p}_j \cdot \mathbf{r}_j)\,\mathbf{r}_j - \mathbf{p}_j |\mathbf{r}_j|^2}{|\mathbf{r}_j|^5} \equiv \sum_{j=1}^{N} \langle s_j \rangle \mathbf{F}_j. \tag{1}$$

The angle brackets indicate a thermal average. In the devices in question, the qubit operation frequency depends linearly on the electric field at its position. The frequency is a quasi-equilibrium quantity, so $\langle \mathbf{F}_q \rangle$ is the object of interest for our purposes. The relation between field and frequency is platform-dependent. In the set-up of Refs. [1, 16, 17], $\langle \mathbf{F}_q \rangle$ causes the displacement of the spin qubit in a magnetic field gradient, while in the flip-flop qubit architecture motion of the qubit caused by $\langle \mathbf{F}_q \rangle$ would change the hyperfine coupling or the g-factor [25]. In all cases the displacement changes the qubit operating frequency. The qubit frequency is $f(T) = f_0 + \mathbf{c}_q \cdot \langle \mathbf{F}_q \rangle (T = 0) + \Delta f(T)$, where f_0 is the T-independent part from the applied magnetic field, $\langle \mathbf{F}_q \rangle (T = 0) \neq 0$ is a constant that comes from the ground state configuration of the TLS, and $\Delta f(T)$ is the PIRS effect and all the T dependence of f comes from it. Thus $\Delta f(T) = \mathbf{c}_q \cdot [\langle \mathbf{F}_q \rangle (T) - \langle \mathbf{F}_q \rangle (T = 0)]$ is the quantity of interest. $\mathbf{c}_q$ depends on the particular type of qubit and the position of the qubit in the device. Its direction is determined by the condition that the effective magnetic field produced by $\langle \mathbf{F}_q \rangle$ should be parallel to the applied field.

The Hamiltonian of the TLS in our model contains a random-field term H_r and an interaction term H_{int}:

$$H = H_r + H_{int} = -p_0 \sum_{j=1}^{N} s_j \mathbf{E}_j \cdot \hat{p}_j - \frac{p_0^2}{8\pi\varepsilon} \sum_{j \neq k=1}^{N} s_j s_k V_{jk}.$$

Here

$$V_{jk} = \frac{3\hat{p}_j \cdot (\mathbf{r}_j - \mathbf{r}_k)\,\hat{p}_k \cdot (\mathbf{r}_j - \mathbf{r}_k) - (\hat{p}_j \cdot \hat{p}_k)\,|\mathbf{r}_j - \mathbf{r}_k|^2}{|\mathbf{r}_j - \mathbf{r}_k|^5}.$$

The random effective electric fields $\mathbf{E}_j$ if interpreted in a double-well picture of the TLS are related to the energy asymmetry ('detuning') of the two wells. However, the physical origin of the $\mathbf{E}_j$ may not be the same in all cases. For example, they could be actual external electric fields coming from the gate electrodes, strain fields, asymmetric microscopic defects, *etc.* For our purposes they are considered to be phenomenological parameters that must be fit, since they are very difficult to estimate in the absence of a real microscopic model. We expect N to be a number in the range of perhaps 10 to 100 and to be sample-dependent [12]. The dipoles may well be the same TLS that give rise to the noise in the system, but here we are interested in their equilibrium behavior, not their fluctuations. This assumes that measurement of PIRS takes place over a time interval longer than the characteristic switching times of the TLS. However, the intersection of the set of TLS that causes qubit dephasing and the set that causes PIRS need not be complete.

III. NON-MONOTONICITY OF PIRS

The T dependence of $\langle \mathbf{F}_q \rangle$ in a non-interacting model with $H_{int} = 0$ is already interesting, so we discuss it in detail. In this case the problem is exactly solvable, once the positions of the dipoles are specified: $\langle s_j \rangle = \mathrm{sgn}(\hat{p}_j \cdot \mathbf{E}_j) \tanh(p_0\, \hat{p}_j \cdot \mathbf{E}_j / k_B T)$. $\langle s_j \rangle$ has a definite sign at $T = 0$ but eventually $\langle s_j \rangle \to 0$ as $T \to \infty$. We can identify a turn-off temperature $T_j = p_0 |\mathbf{E}_j| / k_B$ for each TLS. Substitution into Eq. 1 and performing the sum gives the equilibrium electric field $\langle \mathbf{F}_q \rangle (T)$ at the qubit.

To understand the qualitative T dependence of $\langle \mathbf{F}_q \rangle$ in the IRGM we begin with $T = 0$. We divide the TLS into two groups. In the set S^+ we have the indices j for which $\langle \mathbf{F}_q \rangle (T = 0) \cdot \mathbf{F}_j \langle s_j \rangle (T = 0) > 0$ while in group S^- we have the indices j for which $\langle \mathbf{F}_q \rangle (T = 0) \cdot \mathbf{F}_j \langle s_j \rangle (T = 0) < 0$. That is, the dipoles in S^+ are aligned with the ground state resultant field, while those in S^- are anti-aligned. The electric field at the qubit is the result of a random walk of the vectors $\langle s_j \rangle \mathbf{F}_j$ with a resultant vector

$$\langle \mathbf{F}_q \rangle = \sum_{j \in S^+} \langle s_j \rangle \mathbf{F}_j + \sum_{j \in S^-} \langle s_j \rangle \mathbf{F}_j \tag{2}$$

The vectors in group S^+ are in the direction of the final result of the walk, while those in group S^- are in the opposite direction. Due to the randomness in the asymmetry, the various components of the walk turn off at different temperatures, and there will be some average turn-off temperature T^+ for group S^+ and a different average turn-off temperature T^- for group S^-. We now increase T from zero. If $T^+ > T^-$ then the total field strength $|\langle \mathbf{F}_q \rangle|$ will first increase and eventually vanish when $T \gg T^+$. In this case we have a non-monotonic T-dependence of the field strength $|\langle \mathbf{F}_q \rangle|$, a rather surprising result. T^+ and T^- are not expected to be very different if the dipoles have the same physical structure, but even in this case the relatively small value of N implies that random fluctuations will make $T^+ \neq T^-$. If $T^+ < T^-$ then the total field strength $|\langle \mathbf{F}_q \rangle|$ will decrease as the dominant dipoles turn off and eventually vanish when $T \gg T^-$. If there is a gross mismatch between the T^+ and T^- components of $|\langle \mathbf{F}_q \rangle|$ could even reverse sign, but overall we would expect a monotonic decrease. The relative magnitudes of $|\sum_{j \in S^+} \langle s_j \rangle \mathbf{F}_j|$ and $|\sum_{j \in S^-} \langle s_j \rangle \mathbf{F}_j|$ are also important. If $|\sum_{j \in S^+} \langle s_j \rangle \mathbf{F}_j|$ dominates, then there is little cancellation in the sum and the non-monotonic behavior will be suppressed. If

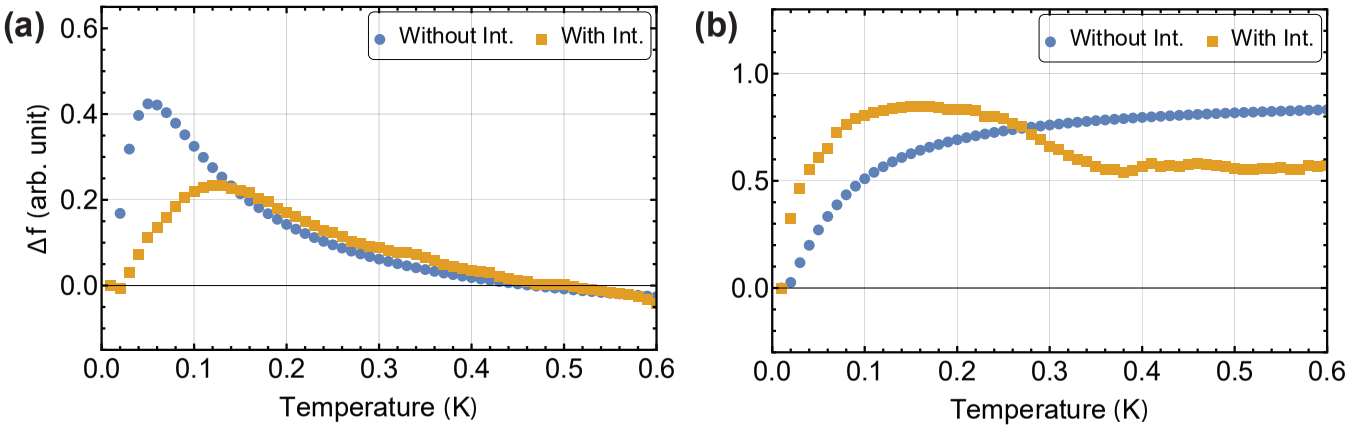


FIG. 1. Temperature dependence of qubit frequency shifts in the trap picture of the TLS in some representative samples. The shifts are proportional to one component of the equilibrium electric field $\langle \mathbf{F}_q \rangle(T)$ at the qubit. Without Int. and With Int. mean the non-interacting and fully interacting case, respectively. **(a)** Both non-interacting and interacting cases show non-monotonic shifts. **(b)** Example in which the interaction causes non-monotonic behavior.

Trap	$\langle F_{q,x} \rangle$	$\langle F_{q,y} \rangle$	$\langle F_{q,z} \rangle$
$T_r < T_{int}$			
Without Int.	15.9%	16.4%	12.0%
With Int.	48.0%	48.6%	40.1%
$T_r \sim T_{int}$			
Without Int.	11.0%	10.9%	7.4%
With Int.	30.5%	30.2%	24.2%

Random dipole	$\langle F_{q,x} \rangle$	$\langle F_{q,y} \rangle$	$\langle F_{q,z} \rangle$
$T_r < T_{int}$			
Without Int.	14.7%	15.9%	12.4%
With Int.	61.9%	61.9%	57.2%
$T_r \sim T_{int}$			
Without Int.	10.0%	9.4%	7.8%
With Int.	42.5%	42.3%	36.9%

Spherical shell	$\langle F_{q,x} \rangle$	$\langle F_{q,y} \rangle$	$\langle F_{q,z} \rangle$
$T_r < T_{int}$			
Without Int.	25.0%	25.3%	28.0%
With Int.	89.5%	89.4%	90.1%
$T_r \sim T_{int}$			
Without Int.	21.0%	21.8%	25.8%
With Int.	79.4%	79.0%	81.4%

TABLE I. The fractional number of samples showing non-monotonicity obtained from Monte Carlo simulation and exact solutions for three physical pictures. $\langle F_{q,i} \rangle$ is the i component of the vector $\langle \mathbf{F}_q \rangle(T)$. Without Int. and With Int. mean the non-interacting and fully interacting case, respectively. $T_r < T_{int}$ indicates $T_r \sim 0.1$ K and $T_{int} \sim 1$ K, while $T_r \sim T_{int}$ indicates $T_r \sim 0.1$ K and $T_{int} \sim 0.1$K.

swer this we did simulations over 10,000 samples for each of the three pictures and determined whether a monotonic or non-monotonic behavior was observed for each component of the electric field. The precise criterion for monotonicity or its absence is given in Appendix C.

The results in Table I support the physical picture explained above. In the trap picture, the cancellation of fields from different dipoles is relatively small, since the vectors leading from the qubit to the TLS, while not collinear, generally do not make large angles with each other. Similar statements apply to the random dipole picture, but there is some additional cancellation due to the different orientations. The most interesting result is for the spherical shell picture. Here we see the appearance of non-monotonic behavior for non-interacting TLS in about one quarter of the cases, as would be suggested by the above arguments. With dipoles on all sides of the qubit, the cancellation effect is quite strong.

Interactions do promote non-monotonicity in all cases, as expected, especially when the interactions are strong: $T_{int} > T_r$. Overall, non-monotonicity increases as we proceed from the trap to the random dipole to the spherical shell pictures. The interaction enhancement reinforces this pattern of non-monotonicity.

We turn now to comparison of theory and experiment. In nearly all measurements Δf is measured as a function of time, not temperature. Comparing to data of this type would require detailed modeling of the heat flow in the system, which is outside the scope of this paper. We therefore compare only to the temperature data for $\Delta f(T)$ for 6 qubits in a single device reported in Ref. [18]. We plot these data with a theoretical fit in Fig. 2. The fit was done as follows. We first chose a set of TLS positions and random fields so that the parameters were in a range where non-monotonicity could be expected, and peak in Δf would be around 250 mK. Then, since the curves have a fairly similar shape but differ in vertical scale, we varied $\mathbf{c}_q$ for each of the curves. Finally, the positions of the TLS were adjusted to fit each curve individually. The main feature that needed to be accounted for in this final step was the sharper peak and steeper falloff at high T that is seen in qubits 3-6. Some further details are given in Appendix D.

The fits are quite good quantitatively, but not too

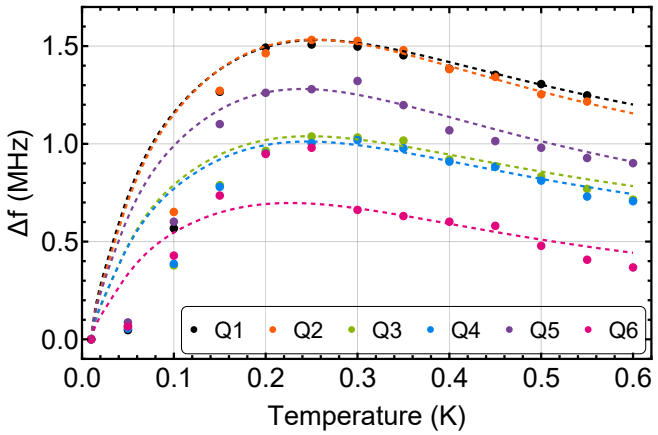


FIG. 2. PIRS data: theory and experiment. The points are measured frequency shifts for six qubits Q1-Q6 from Ref. [18] and the dashed lines are theoretical fits. The qubits are situated in a one-dimensional array. The fitting procedure is described in detail in the text. The applied magnetic field is assumed to be in the y-direction.

much should be made of this, since the number of parameters far exceeds the number of qualitative features to be fit for each curve. However, even given this, the fit does provide evidence for the correctness of the IRGM. The non-monotonicity arises naturally, but also the linear behavior at small T, (which is caused by the uniform distribution of the $\mathbf{E}_j$ near zero) and finally the $1/T$ behavior at large T.

V. CONCLUSION

We conclude by summarizing the strong and weak points of the IRGM as a model for PIRS. There are three evaluation categories; qualitative phenomenological understanding, semi-quantitative self-consistency, and quantitative fit of theory and experiment.

In qualitative terms, the surprising non-monotonic T dependence and some of the other features of the T-dependence of the qubit frequencies finds a natural explanation in the IRGM. The important theoretical ingredients are the cancellations due to vector summations that involve only a relatively small number of variables combined with the natural temperature dependence of the TLS fluctuations and the depolarizating effects of interactions. The explanation of non-monotonicity is quite defies the usual expectation that thermal effects in the absence of phase transitions tend to be monotonic. Overall, this is quite strong evidence for the IRGM. Aside from the non-monotonicity, there is the observation that Δf is sometimes negative [16, 19]. This is also somewhat surprising if one assumes that the heating affects the qubits directly in some fashion. In the IRGM, there is nothing that constrains the sign of the components of $\mathbf{c}_q$, so the sign of the effect is not determined. Similarly, the fact that the effect is not resonant with qubit frequencies suggests that an ancillary part of the device is driven by the heating - in the IRGM, the system of TLS is driven. Hysteresis does not seem to be a feature of PIRS. This might seem to argue against the IRGM but in fact with only a few tens of TLS involved, this aspect of glassiness does not argue against the model. In contrast, no T-dependent electric field shows up in measurements at the charge sensor [18], which is not explained in the model as it stands.

There are two experimental scales that must be consistent with the theory: the overall magnitude of Δf (1 MHz in dot systems), and the temperature of the peak in Δf (about 0.2 to 0.4 K in dot systems). The first number is very consistent with the roughly known numbers for the size of the dipoles and their presumed positions. The second depends on the distribution of the random local fields $\mathbf{E}_j$ and indeed the distribution must be such that $|p_0\mathbf{E}_j/k_B|$ is clustered near 0.3 K. There is no obvious reason why this should be so, so the IRGM does include at least one *ad hoc* element.

The fit to the data in Fig. 2 is strikingly accurate, but it raises questions. If interactions are relatively weak for some reason (such that the TLS are particularly far apart), then why do six out of six qubits all show non-monotonicity? This is only consistent with the spherical shell picture, which in turn is not very consistent with the usual idea that the TLS are associated with the oxide layer. Furthermore, the non-monotonicity in all six qubits means that there must be correlations between the positions of the TLS and the strength and direction of their random fields. Specifically, the TLS closer to the qubits must have stronger random fields for each qubit. In addition, in order for the shift to be positive in all qubits, the random fields for the nearby TLS must all have the same orientation across all six qubits.

We conclude that the basic mechanism of PIRS is explained by the IRGM, but that the explanation is far from complete at this stage. Most likely the model needs to supplemented by a better picture of the positions of the TLS and a better understanding of their physical nature. This would limit the model to a smaller region of its parameter space than we have investigated here.

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